IMPERIAL COLLEGE LONDON

EE4-26
EE9-CS5-3
EE9-FPN2-07

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2020

MSc and EEE PART IV: MEng and ACGI

ESTIMATION AND FAULT DETECTION

Tuesday, 19 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : T. Parisini
Second Marker(s) : D. Angeli

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ESTIMATION AND FAULT DETECTION

Recommendations for candidates:

The exam has to be carried out remotely following the official indications provided by the College. Moreover, the algebraic calculations have to be carried out using a standard calculator. The use of mathematical software such as Matlab and similar tools installed on personal computers is prohibited.

Information for candidates:

- **One-step ahead Kalman predictor**:

$$\hat{x}(t+1|t) = F\hat{x}(t|t-1) + K(t)[y(t) - H\hat{x}(t|t-1)]$$

- **Kalman predictor gain**

$$K(t) = FP(t)H^\top \left(V_2 + HP(t)H^\top\right)^{-1}, \quad t = 1,2,\dots$$

- **Riccati equation**

$$P(t+1) = F\left[P(t) - P(t)H^\top \left(V_2 + HP(t)H^\top\right)^{-1} HP(t)\right]F^\top + V_1, \quad t = 1,2,\dots$$

- **Realization: observer canonical form**

  Given

$$Y(s)/U(s) = \frac{b_m s^m + \cdots + b_1 s + b_0}{s^n + a_{n-1}s^{n-1} + \cdots a_1 s + a_0} \quad \text{with} \quad m < n$$

  then:

$$\begin{cases} \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_{n-1} \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} 0 & 0 & \cdots & 0 & -a_0 \\ 1 & 0 & \cdots & 0 & -a_1 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & \cdots & 1 & 0 & -a_{n-2} \\ 0 & \cdots & 0 & 1 & -a_{n-1} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} + \begin{bmatrix} b_0 \\ b_1 \\ \vdots \\ b_m \\ 0 \\ \vdots \\ 0 \end{bmatrix} u \\ \\ y = [0 \cdots 0\,1] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} \end{cases}$$

- **Generic Structure of a Reduced-Order Observer**

$$\begin{cases} \dot{z}_1 = \tilde{A}_{11}z_1 + \tilde{A}_{12}z_2 + \tilde{B}_1 u \\ \dot{z}_2 = \tilde{A}_{21}z_1 + \tilde{A}_{22}z_2 + \tilde{B}_2 u \\ v = z_1 \end{cases} ; \quad \begin{cases} \dot{z} = Fz + Hv + Gu \\ \widehat{x}_2 = z + Lv \end{cases} ; \quad \begin{cases} L\tilde{A}_{12} - \tilde{A}_{22} = -F \\ L\tilde{A}_{11} + H - \tilde{A}_{21} = FL \\ G + L\tilde{B}_1 - \tilde{B}_2 = 0 \end{cases}$$

1. Consider a one-link flexible joint manipulator actuated by a DC motor, as shown in Figure 1.1:

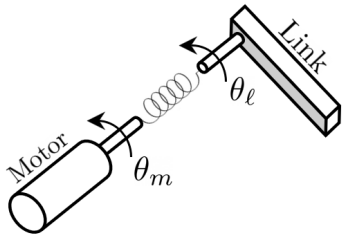


Figure 1.1 A one-link flexible joint manipulator.

The elasticity of the joint is modelled by a linear torsional spring. The state variables of this system are chosen to be the motor angular position, the link angular position, and the link angular velocity. Accordingly, the state-space model is given by:

$$\begin{cases} \dot{\theta}_m = u \\ \dot{\theta}_\ell = \omega_\ell \\ J_\ell \dot{\omega}_\ell = -K(\theta_\ell - \theta_m) + MgL\theta_\ell \end{cases} \tag{1.1}$$

where $J_\ell = 1$ is the inertia of the link, θ_m and θ_ℓ are the angular rotations of the motor and the link, respectively, ω_ℓ is the angular velocity of the link, and u is the control input. Moreover, $M = 1$ is the mass of the link, g is the gravity acceleration, $L = 1$ is the length of the link, and $K > g$ is the stiffness coefficient of the torsional spring. A sensor is available providing an exact measurement of the angular rotation of the motor, that is:

$$y = \theta_m \tag{1.2}$$

a) Show that the system is not fully observable. Subsequently, determine a basis for the non-observable subspace.

[5 Marks]

b) Set $u = 0$. Find a non-zero initial state $[\theta_m(0), \theta_\ell(0), \omega_\ell(0)]^\top$ giving rise to an identically-zero output, that is $y(t) = 0, \forall t \geq 0$.

[3 Marks]

c) Determine the state equations of the observable and the non-observable subsystems of System (1.1).

[5 Marks]

d) Obtain the non-observable eigenvalues of the system.
Hint: Make an appropriate use of your answer to Question 1-c).

[3 Marks]

e) Discuss the addition of a sensor providing the measurement of another state variable (in addition to θ_m in Equation (1.2)) so as to make System (1.1) (possibly) fully observable.

[4 Marks]

2. Consider the discrete-time dynamical system described in Figure 2.1 where $d_1(\cdot) \sim WGN(0,1)$ and $d_2(\cdot) \sim WGN(0,3)$ are stochastic processes which are supposed to be independent from each other.

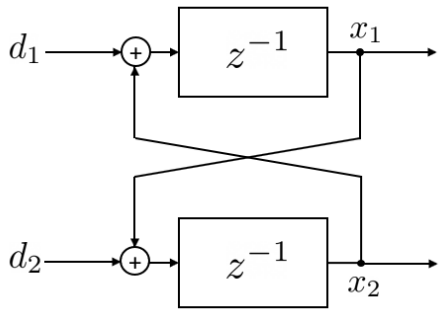


Figure 2.1 Block-scheme of the system of Question 2.

a) Determine a stochastic state-space representation of the system sketched in Figure 2.1, with state defined as $x(t) = [x_1(t), x_2(t)]^\top$.

[3 Marks]

b) By employing a suitable similarity transformation matrix $T \in \Re^{n \times n}$, transform the state-space representation given in your answer to Question 2-a) to an equivalent state-space representation, with state $z(t) = Tx(t)$, such that the equivalent state matrix is diagonal.

[3 Marks]

c) Consider the state-space representation obtained in your answer to Question 2-b). Assume that sensors have been used to provide the output

$$y(t) = z(t) + \xi(t)$$

in which $\xi(t) = [\xi_1(t), \xi_2(t)]^\top$, with $\xi_1(\cdot) \sim WGN(0,1)$ and $\xi_2(\cdot) \sim WGN(0,2)$ being stochastic processes supposed to be independent from each other and from $d_1(\cdot)$ and $d_2(\cdot)$. Consider the one-step ahead optimal steady-state Kalman predictor of the state $z(t)$. Show that the Algebraic Riccati Equation (ARE) admits a feasible matrix solution P and compute it.
Hint: Since $d_1(\cdot)$ and $d_2(\cdot)$ are independent, independent disturbances $\bar{d}_1(t) = \frac{1}{2}d_1(t) + \frac{1}{2}d_2(t)$ and $\bar{d}_2(t) = -\frac{1}{2}d_1(t) + \frac{1}{2}d_2(t)$ can be defined.

[5 Marks]

d) Compute the gain vector K associated with the ARE of your answer to Question 2-c) and write the difference equation yielding the steady-state Kalman prediction $\hat{z}(t|t-1)$.

[4 Marks]

e) Compute the covariance matrix $\mathrm{cov}[z(t) - \hat{z}(t|t-1)]$ of the prediction error $z(t) - \hat{z}(t|t-1)$.

[5 Marks]

3. Consider a continuous-time dynamical system described by the following state equations:

$$\begin{cases} \dot{x}_1(t) = (3-\alpha)x_1(t) + (1+\alpha)x_2(t) + u(t) \\ \dot{x}_1(t) = 2x_1(t) + \alpha x_2(t) + u(t) \\ y(t) = x_1(t). \end{cases} \tag{3.1}$$

a) Find the set of **all** $\alpha \in \Re$ for which the system is observable.

[3 Marks]

b) Set $\alpha = 1$. By defining $\hat{x}(t)$ as the estimate of $x(t) = [x_1(t), x_2(t)]^\top$ and by letting $e(t) = x(t) - \hat{x}(t)$, design a full-order state observer for which the estimation error has the following dynamics

$$\dot{e}(t) = Fe(t)$$

where F has eigenvalues at -1 and -2.

[4 Marks]

c) Set $\alpha = -1$. Is it possible to design a full-order state observer under which the estimation error has the following dynamics:

$$\dot{e}(t) = Fe(t)$$

where F has eigenvalues at -1 and -2? Discuss your findings.

[4 Marks]

d) Set $\alpha = 1$. Assume that the control input is generated by the algebraic output feedback law

$$u(t) = Ky(t) + f \quad t \geq 0, \tag{3.2}$$

where $K \in \Re$ is a given feedback gain and f denotes a **constant** fault on the actuator. By considering the same observer you have designed in your answer to Question 3-b), determine the differential equation describing the dynamics of the estimation error $e(t) = x(t) - \hat{x}(t)$ in the presence of the actuator fault f. Then, obtain the expression of the output residual $\varepsilon(t) = Ce(t)$ over time where $C = [1 \;\; 0]$.

[3 Marks]

e) Referring to your answer to Question 3-d), assume that the initial conditions satisfy $\|e(0)\| \leq 1$. Find the set of all the faults f which can be detected with a threshold of 1 at $t = 1$. In other words, for which f do we have $|\varepsilon(1)| > 1$?

[3 Marks]

f) Set $\alpha = 1$. Assume that the control input is given by the control law (3.2). Show that an observer-based scheme can be devised to estimate f.

[3 Marks]

4. Consider a continuous-time dynamical system described by the following state equations:

$$\begin{cases} \dot{x}_1(t) = -x_2(t) \\ \dot{x}_2(t) = 2x_3(t) + x_2(t) \\ \dot{x}_3(t) = -x_2(t) + x_3(t) \\ y(t) = x_1(t) + cx_3(t) \end{cases} \tag{4.1}$$

where $c \in \mathfrak{R}$ is a constant.

a) Find the set of all $c \in \mathfrak{R}$ for which the system is fully observable.

[4 Marks]

b) Determine the order of the reduced-order state observer for the system. Discuss your findings.

[3 Marks]

c) Set $c = 0$. Design a reduced-order state observer for the unmeasured state vector $x_r(t) = [x_2(t), x_3(t)]^\top$ such that, by defining $\hat{x}_r(t)$ as the estimate of $x_r(t)$, the estimation error $e(t) = x_r(t) - \hat{x}_r(t)$ obeys the following dynamics:

$$\dot{e} = Fe$$

where all the eigenvalues of F are -1 (more than one solution is available).

[7 Marks]

d) Set $c = 0$. Assume that a **constant** fault f affects the system (4.1) yielding the new state equations

$$\begin{cases} \dot{x}_1(t) = -x_2(t) + f \\ \dot{x}_2(t) = 2x_3(t) - 3x_2(t) \\ \dot{x}_3(t) = -x_2(t) - 2x_3(t) \\ y(t) = x_1(t). \end{cases} \tag{4.2}$$

Is there any **constant** fault f such that, for every initial state, the corresponding state evolution $x(t)$ is **unbounded**. Discuss your findings.

[6 Marks]